WASP-52b

R-0464

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_220918` · created 2026-07-19T21:58:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459840.9342 +/- 0.0069 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.064
Transit depth (Rp/Rs)^2	3.13 +/- 2.25 [%]
Orbital Inclination (inc)	81.81 +/- 2.77
Airmass coefficient 1 (a1)	0.57 +/- 0.32
Airmass coefficient 2 (a2)	0.33 +/- 0.14
Scatter in the residuals of the lightcurve fit is	58.75 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	3.5
Transit Duration (day)	0.038 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.177 ± 0.064 vs 0.1646 (deviation 0.19σ)
Duration fit vs published	0.038 d vs 0.0758 d (deviation 1.15σ)
Mid-time O–C	12.7 min
Depth SNR	2.8
Residual scatter	58.75 %

Light curve

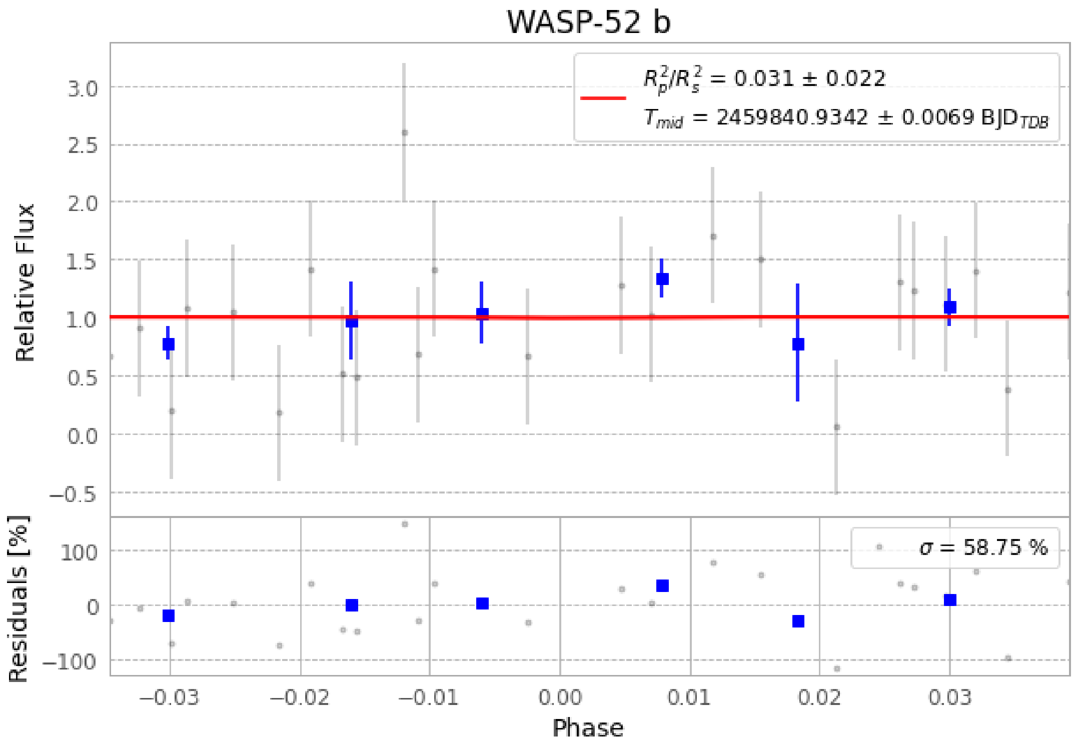

Figure 1 — FinalLightCurve_WASP-52 b_2022-09-18.png (SHA-256 cae81723db4beb32…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 132], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6feebdaf0b722931…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

64 FITS frames (42 MB), `WASP-52220918084516.FITS` → `WASP-52220918115415.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-220918.log (37651aeade4e…), FinalLightCurve_WASP-52 b_2022-09-18.png (cae81723db4b…), FinalLightCurve_WASP-52 b_2022-09-18.csv (e724527b47f5…)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)